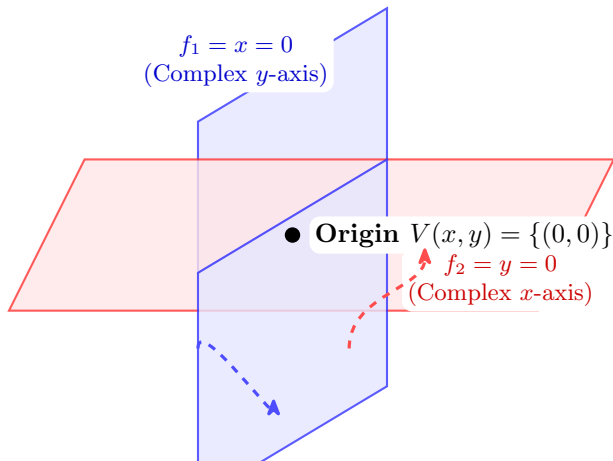




Defining the Syzygy over $\mathbb{C}[x, y]$

Generator $\mathbf{f}_1 = \mathbf{x}$: Vanishes entirely on the vertical complex plane.

Generator $\mathbf{f}_2 = \mathbf{y}$: Vanishes entirely on the horizontal complex plane.

The Hidden Constraint (Syzygy):

To create a combination that vanishes *everywhere* in $\mathbb{C}^2$, we cross-multiply the generators by each other's defining functions:

$$(-y) \cdot \mathbf{f}_1 + (x) \cdot \mathbf{f}_2 = (-y)x + (x)y = \mathbf{0}$$